

[NJC/2025/Q3 Cont ']

Task 3.4

Write a Python function `search_birthday(sorted_birthdays, date)` that uses the **binary search** algorithm to find the smallest and largest indices (inclusive) in the list of `Person` objects whose birthdate matches the given date, where

- `sorted_birthdays` is the sorted list of `Person` objects returned by the function `sort_birthdays()` in Task 3.2.
- `date` is the target to be searched, a date string in the format **YYYYMMDD**.
- returns the leftmost(smallest) and rightmost(largest) indices of the elements in the list whose birthdate matches `date`.

Example, calling `search_birthday(sorted_birthdays, "19980701")` should return `(53, 56)` indicating that the elements at indices 53, 54, 55, 56 have birthdate that matches `date`.

- returns `(-1, -1)` if there are no matches.

Note:

- there may be more than one person with the same birthday.
- no marks will be given if the solution does not use a binary search algorithm. Full marks for this sub-task will only be given if the solution has a worst-case time complexity of $O(\log N)$.

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Task 3.5

Write Python code that uses the indices returned by the function

`search_birthdays` to print a list of matching `Person` objects from

`sorted_birthdays`.

Test Cases	Expected Output
<code>search_birthdays</code> <code>(sorted_birthdays, "19980701")</code>	<code>[<Alice Smith:19980701>, <Sarah Smith:19980701>, <Eric Banner:19980701>, <Emily Brown:19980701>]</code>
<code>search_birthdays</code> <code>(sorted_birthdays, "19841231")</code>	<code>[<Katie Brown:19841231>]</code>
<code>search_birthdays</code> <code>(sorted_birthdays, "20041231")</code>	<code>[]</code>

[2]

Save your Jupyter Notebook for Task 3.